

Spacesuit Thermal Insulation Laboratory

Development and validation of a classroom thermal protection experiment



The complete four-configuration laboratory system prepared for testing.

Engineering Project Report

Human Spaceflight Technology

TUM School of Engineering and Design

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Executive Summary

This report documents the current SPHERE thermal-insulation laboratory: a controlled comparison of four 100 mL water capsules exposed to a cold water-bath boundary. The final system combines physical test hardware, an interactive browser application, manual temperature logging, Newtonian cooling predictions, and post-test error analysis. The updated report replaces draft PET-container and unverified performance claims with the settings, photographs, model coefficients, and worked telemetry contained in the current project files.

The test uses a bare control, bubble wrap, reflective Mylar, and a full multilayer package consisting of cotton, bubble wrap, and Mylar. Each run records the core temperature at $t = 0, 1, \dots, 15$ minutes. The website generates a reference curve and overlays manually entered measurements. The supplied test dataset shows the full MLI package retaining the most heat, while Mylar alone performed poorly in the liquid-bath environment. These observations are illustrative and depend on probe position, sealing, layer compression, bath temperature, and starting conditions.

Verified Project Settings

Setting	Current project value	Reason
Thermal core	100 mL wide-mouth glass jar	Consistent geometry and thermal mass
Initial temperature	Approximately 80 °C; record actual T_0	Reproducible but never boiling
Boundary	Measured ice-water bath; nominal 0 °C	Enter actual T_{env} if the bath warms
Sampling	Every minute from 0 through 15 min	16 time-temperature pairs
Conditions	Bare, bubble wrap, Mylar-only, full MLI	Controlled comparison
Analysis	Temperature drop, fitted k , residuals and MAE	Separates observation from prediction

Contribution Attribution

For this updated submission, Piyath Vithanage is credited with website development and graphing/data visualisation. Dilan Fernando is credited with thermodynamics and the remaining project work, including the experimental system, analysis, documentation, and integration. The full four-person project team remains listed because those names are present in the current repository and learning materials.

1 Project Context and Objectives

The SPHERE project translates human-spaceflight engineering into accessible classroom activities. This module uses the thermal-control challenge of a spacesuit as the narrative frame for a measurable terrestrial experiment. The aim is not to reproduce vacuum; it is to let learners build, measure, model, and critique a simplified thermal system.

1.1 Learning Objectives

- Distinguish conduction, convection, and thermal radiation in a practical system.
- Apply Newton's law of cooling and interpret an effective cooling constant k .
- Collect a controlled temperature time series and compare it with a prediction.
- Use residuals and mean absolute error to evaluate model agreement.
- Explain how uncontrolled variables can change an apparent insulation ranking.

1.2 Scope and Constraints

The activity is designed for four groups working in parallel. The expected class duration is approximately 55–70 minutes, including setup, the 15-minute measurement interval, and post-processing. The equipment is reusable and relies on a web browser rather than a native application or wireless sensor link.



Theory and safety preparation before the physical test.

2 Thermodynamic Framework

The warm water is treated as a lumped thermal mass. Its stored sensible energy change is estimated from $Q = m c_p \Delta T$, where m is the water mass, c_p is the specific heat capacity of water, and ΔT is the measured temperature change. For 100 mL of water, the mass is approximately 0.10 kg, subject to the actual fill measurement.

2.1 Newtonian Cooling Model

$$T(t) = T_{env} + (T_0 - T_{env}) e^{(-kt)}$$

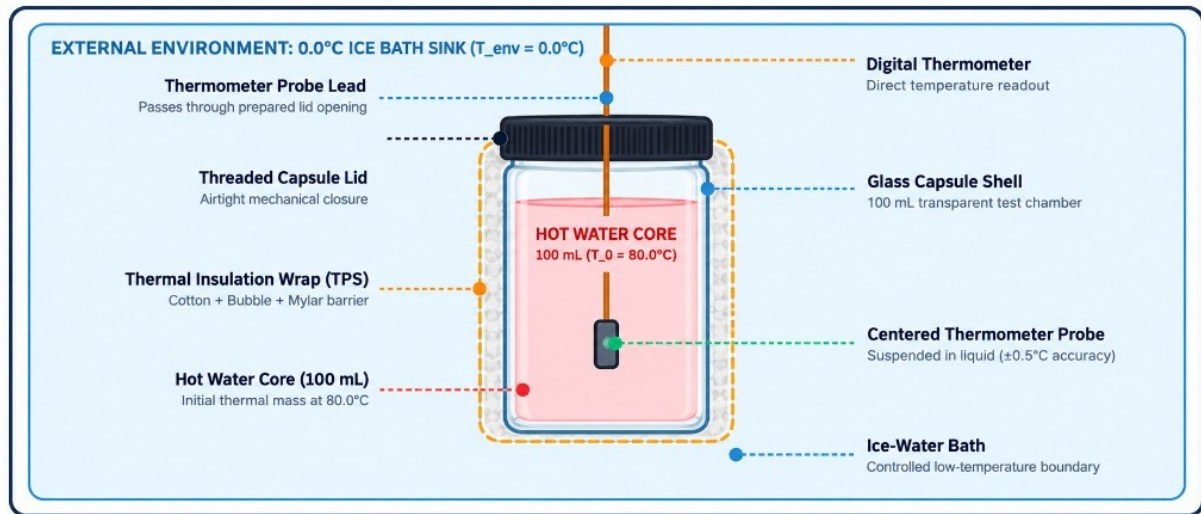
The effective cooling constant k has units of min^{-1} . It combines geometry, thermal mass, boundary conditions, and all heat-transfer paths; it is not a material thermal conductivity. A smaller k indicates slower approach to the bath temperature under comparable conditions.

2.2 Heat Transfer Interpretation

Mechanism	Classroom system	Spacesuit connection
Conduction	Direct heat flow through the jar, water, and solid layers	Layered fabrics and structural contacts create conductive paths.
Convection	Water motion in the bath and air movement within gaps	External convection is absent in vacuum, but internal garment ventilation is important.
Radiation	Infrared exchange among warm surfaces and surroundings	Reflective low-emissivity layers reduce radiative transfer in vacuum.

The ice-water bath is primarily a conductive and convective environment. Consequently, a reflective sheet can perform poorly when used alone, especially if liquid contact or an open neck provides a stronger heat-loss path. The experiment is an analogy, not a vacuum qualification test.

3 Hardware and Test Configurations



Current hardware arrangement: glass core, centred probe, closure, insulation, and external water bath.

3.1 Apparatus

Quantity	Item	Function
4	Identical 100 mL glass jars with probe access	Parallel thermal cores
4	Digital probe thermometers	Core temperature measurement
1+	Separate bath thermometer	Boundary calibration
1	Stable bucket or deep container	Cold bath and secondary containment
1 set	Cotton, bubble wrap, Mylar, tape or bands	Insulation configurations
1	Timer	One-minute sampling

3.2 Layer Construction

Condition	Construction	Reference k
Bare control	Uninsulated jar with consistent probe depth	0.150 min^{-1}
Bubble wrap	Two even layers; air cells uncrushed	0.040 min^{-1}
Mylar only	One reflective layer; neck closed	0.121 min^{-1}
Full MLI	Cotton inner, bubble wrap middle, Mylar outer	0.015 min^{-1}

4 Experimental Method

4.1 Controlled Procedure

Step	Action
1	Prepare a stable dry area and separate laptops and displays from water.
2	Measure and record the actual bath temperature T_{env} .
3	Fill each thermal core with 100 mL of water near 80 °C and record its actual T_0 .
4	Centre the probe at a repeatable depth and check the closure for leaks.
5	Apply the assigned insulation without compressing the cotton or bubble cells.
6	Place the core in the bath, start the timer immediately, and record $t = 0$.
7	Record the core temperature every minute through $t = 15$ minutes.
8	Calculate temperature drop, fit or select k , compute residuals and MAE, and discuss uncertainty.

4.2 Fair-Test Requirements

All conditions should use the same jar geometry, water volume, probe depth, sampling interval, bath location, and starting-temperature tolerance. When starting temperatures differ, rank designs using temperature drop, normalised heat retained above the measured boundary, or a fitted cooling constant rather than final temperature alone.



Physical setup immediately before immersion and timed measurement.

5 Physical Configuration Images



Bare control



Bubble wrap



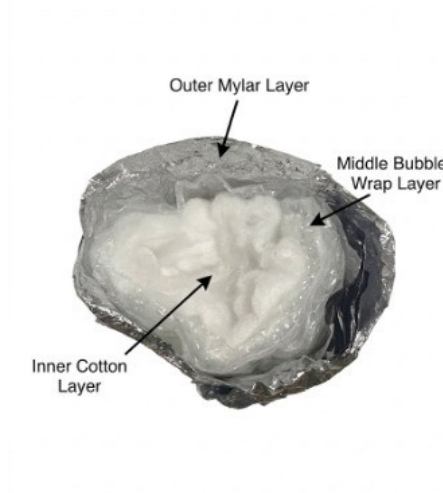
Mylar only



Full MLI

5.1 Full MLI Stack

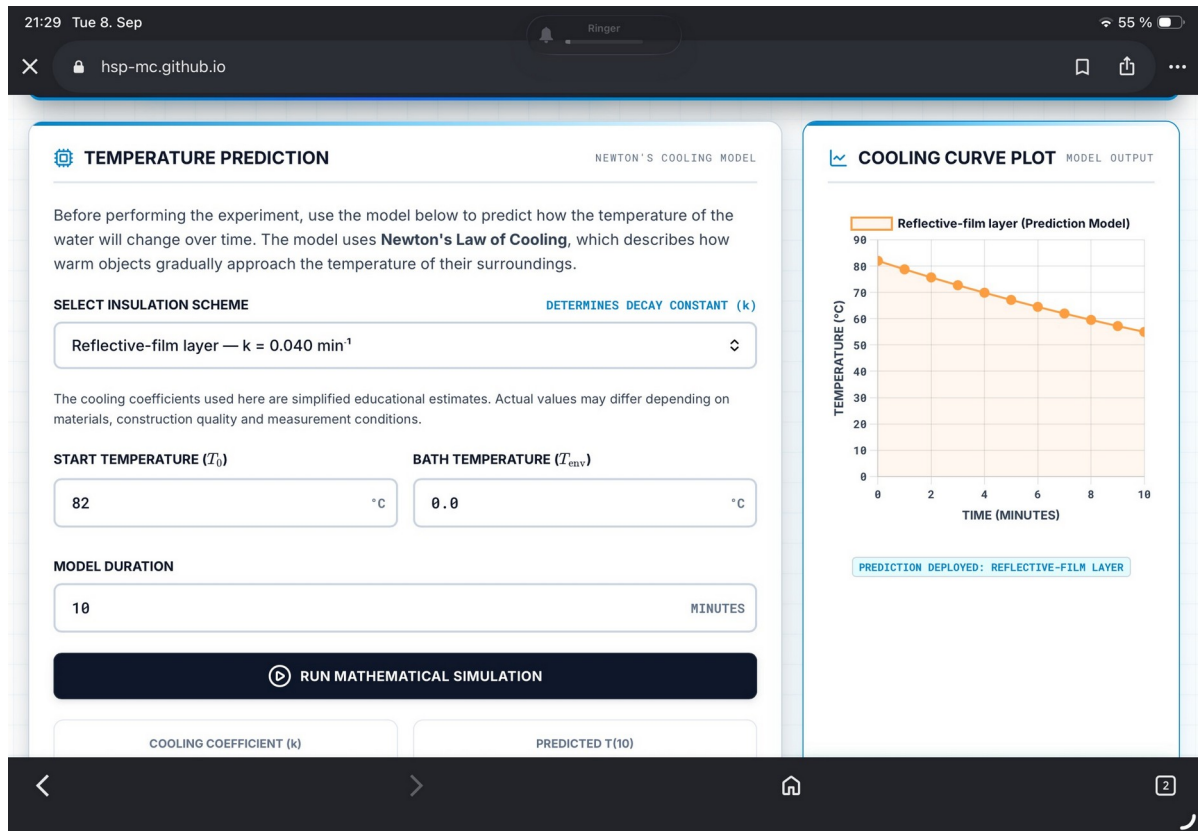
The full package is assembled from the core outward: a uniform cotton layer, two layers of bubble wrap, and a reflective Mylar outer layer. Each layer is secured independently to avoid crushing trapped air. Probe position and neck closure must remain consistent across all tests.



Cross-section of the full multilayer insulation package.

6 Website and Data Workflow

The Mission Control website is a client-side application built with HTML, CSS, and JavaScript. It runs in a standard browser, provides the theory and assembly sequence, calculates reference curves, accepts manual temperature entries, and plots measured values against the model. Chart.js is used for dynamic visualisation; KaTeX supports equations; local browser storage preserves the active prediction and telemetry.



Mission Control prediction interface with editable start temperature, bath temperature, model duration, and cooling constant.

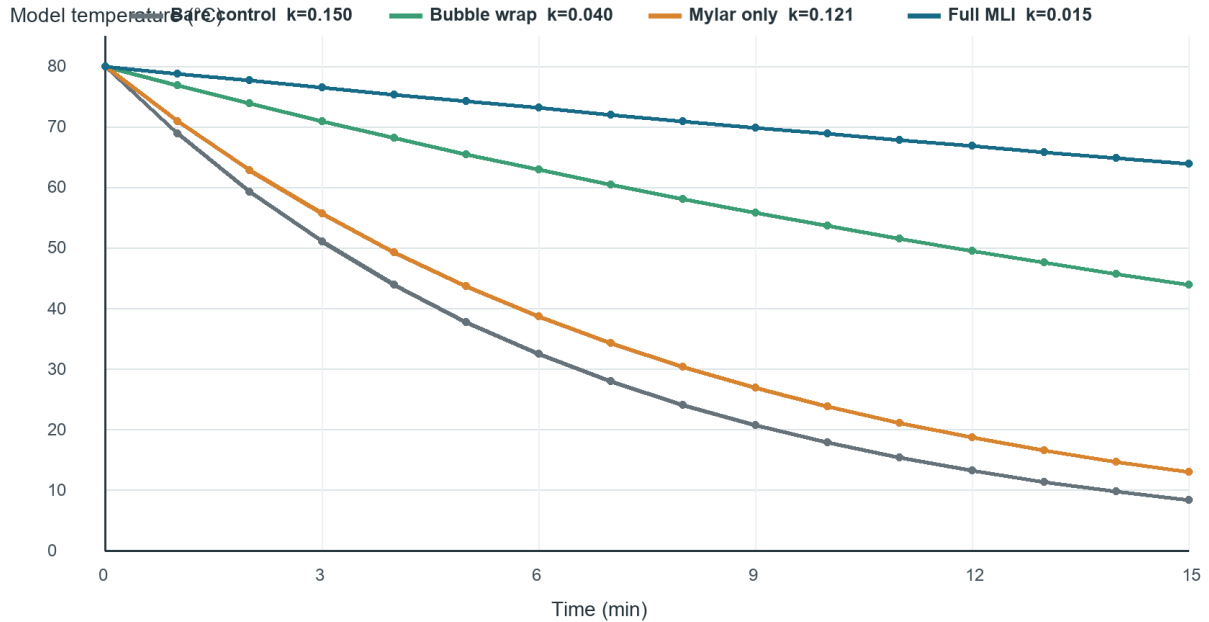
6.1 Measurement and Analysis Flow

Stage	Website function	Student action
Prediction	Generate T(t) from T ₀ , T _{env} , k, and duration	Select a configuration and record assumptions.
Acquisition	Timer and 16-point manual log	Read the physical display every minute.
Comparison	Overlay measurement and prediction	Inspect residuals and curve shape.
Evaluation	MAE and correlation-style feedback	Explain disagreement using physical mechanisms.

Contribution note. Website development and graphing/data visualisation are attributed to Piyath Vithanage in this updated report.

7 Reference Model Results

The website ships with educational reference coefficients for a common starting point of $T_0 = 80\text{ °C}$ and $T_{env} = 0\text{ °C}$. These coefficients are model settings, not guaranteed material properties or validated universal values. They must be adjusted when geometry, water mass, bath temperature, or construction changes.

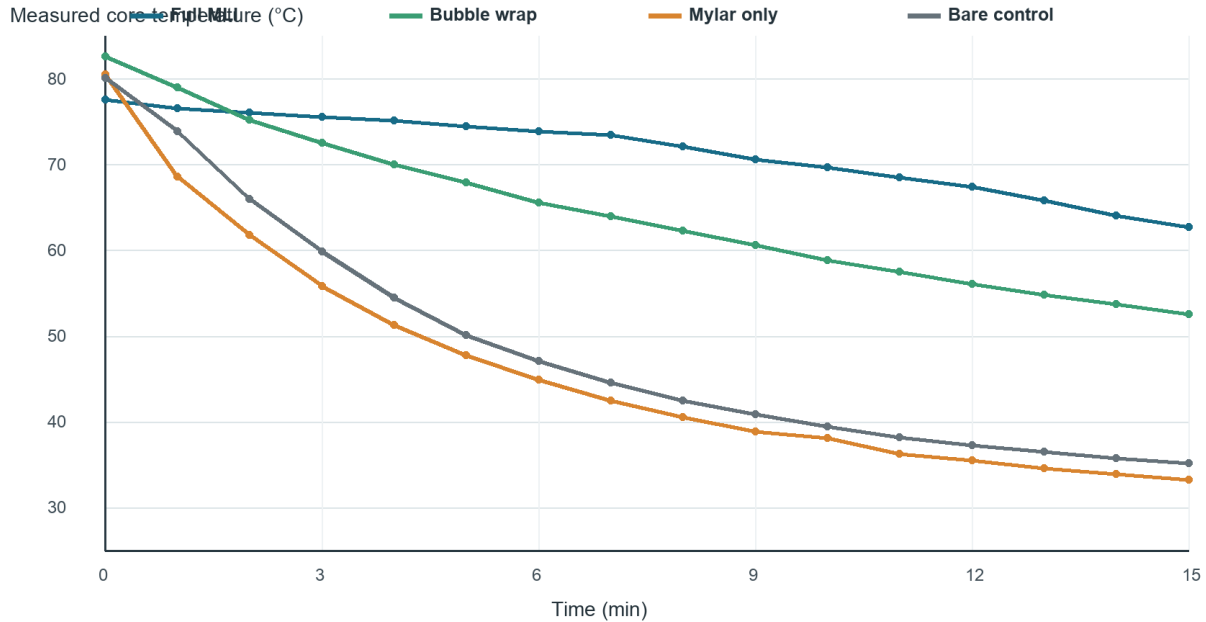


Reference Newtonian cooling curves generated from the current website coefficients.

Configuration	k (min ⁻¹)	Model T15 at 80/0 °C	Interpretation
Bare control	0.150	8.4 °C	Fast reference decay
Bubble wrap	0.040	43.9 °C	Moderate retention
Mylar only	0.121	13.0 °C	Weak alone in this bath model
Full MLI	0.015	63.9 °C	Slowest model decay

8 Worked Experimental Results

The following dataset was transcribed from the current project's telemetry records. It is retained as a worked example and should not replace repeated student measurements. Because T_0 differs among trials, final temperature alone is not a fully controlled ranking metric.



Observed temperature histories from the supplied four-condition test.

Condition	T_0	T_{15}	Drop	Dashboard score
Full MLI	77.6 °C	62.7 °C	14.9 °C	93%
Bubble wrap	82.6 °C	52.6 °C	30.0 °C	83%
Bare control	80.1 °C	35.2 °C	44.9 °C	94%
Mylar only	80.5 °C	33.3 °C	47.2 °C	3%

The full MLI package showed the smallest measured drop. Bubble wrap was second. Mylar-only cooled slightly more than the bare control in this run; the low dashboard score indicates that the observed curve strongly disagreed with its selected reference model.

9 Analysis and Model Limitations

9.1 Error Metrics

$$\text{Residual}_i = T_{\text{measured},i} - T_{\text{model},i} \quad | \quad \text{MAE} = (1/n) \sum |\text{Residual}_i|$$

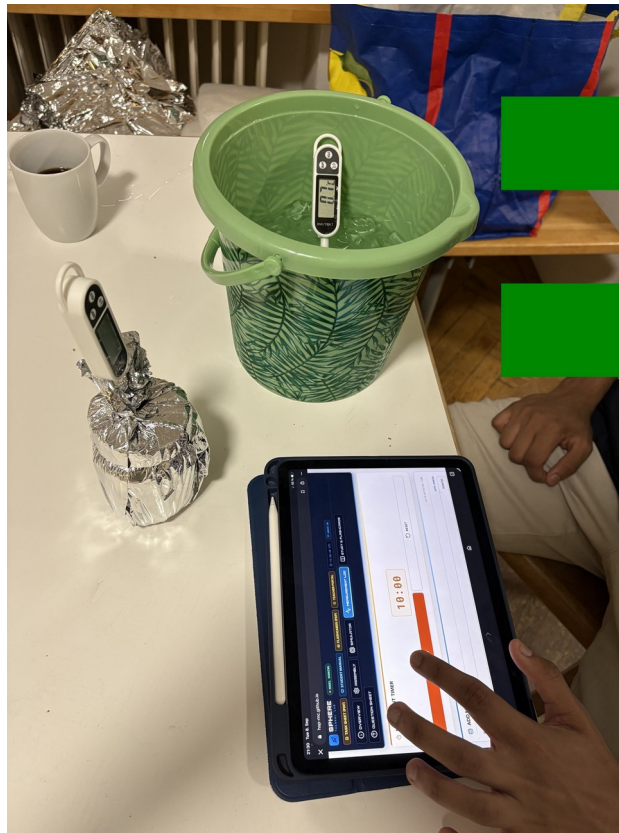
MAE reports the mean absolute temperature difference over all recorded points. A fitted cooling constant may be estimated from $k = -\ln[(T(t) - T_{\text{env}})/(T_0 - T_{\text{env}})] / t$ when the ratio is positive and the boundary is approximately constant.

9.2 Main Sources of Uncertainty

- Probe tip touching the jar wall instead of remaining centred in the water.
- Leakage or an incompletely closed neck that creates a strong convective path.
- Thermal stratification in the core and inconsistent mixing between trials.
- Ice-bath warming above 0 °C or different immersion depths.
- Different starting temperatures, water masses, layer areas, or compression.
- The lumped model's assumption of a uniform core temperature and constant effective k .

9.3 Interpretation of Mylar

Mylar reflects infrared radiation, which is important in space vacuum. A single thin reflective layer does not block direct liquid conduction and can be defeated by convective leakage. The weak Mylar-only result therefore demonstrates a boundary-condition and construction issue; it does not establish that reflective insulation is ineffective in spacecraft thermal control.



Manual entry during the timed physical run keeps measurement and model comparison linked.

10 Safety and Feasibility

10.1 Mandatory Safety Controls

- An instructor or adult prepares and handles water near 80 °C; boiling water is prohibited.
- Wear eye protection and heat-resistant gloves or use tongs. Keep the bath below waist height on a stable surface.
- Use only undamaged, heat-safe containers. Leak-test the closure before immersion.
- Keep displays, connectors, laptops, and power supplies dry; only the intended probe enters water.
- Stop if a probe slips, insulation wets through, the container leaks, or T_0 is outside the agreed tolerance.

10.2 Current Bill of Materials

Item	Quantity	Function
100 mL glass thermal cores	4	Comparable test vessels
Digital probe thermometers	4 + bath probe	Core and boundary measurement
Bucket or deep container	1	Cold bath and containment
Cotton, bubble wrap, Mylar	1 set	Layer configurations
Tape or elastic bands	As needed	Repeatable assembly
Gloves/tongs, towels, timer	1 set	Safe handling and timing

The hardware is reusable, and the browser application requires no installation. Exact cost depends on local sourcing; the older €35.20 estimate should be treated as a draft procurement figure rather than a verified current price.

11 Contributions and Project Integration

Contributor	Updated attribution
Piyath Vithanage	Website development; interface integration; model and telemetry graphs; data visualisation.
Dilan Fernando	Thermodynamic model; experimental design and settings; hardware/test integration; analysis; documentation; remaining project work.
Elio Krollpfeiffer	Listed as a project team member in the current repository and learning materials.
Nuhansee Migelhewage	Listed as a project team member in the current repository and learning materials.

11.1 Integration Outcome

The project's strongest feature is the closed loop between theory and observation: learners choose a configuration, predict its response, build the physical package, acquire a time series, and evaluate disagreement. The website does not automate the thermometer; manual entry preserves the experimental act of observation while still providing immediate graphs and quantitative feedback.

12 Conclusion

The updated SPHERE thermal-insulation laboratory is a coherent, reusable teaching system for transient heat transfer and engineering validation. Its correct project settings are now aligned across the report, website, manuals, SOP, and photographs. The full MLI worked example showed the best heat retention, but the report clearly separates model settings from measured evidence and identifies the limits of the terrestrial water-bath analogy.

References and Project Sources

- SPHERE Mission Control website: index.html, app.js, and styles.css.
- SPHERE student manual, teacher manual, task sheet, and flashcards in the project repository.
- Experimental Toolkit Procedure: Thermal Insulation and Cooling Rates for Spacecraft Thermal Protection, version dated 24 July 2026.
- Current project photographs and diagrams in the images directory.

Safety note. This classroom experiment does not certify any material for aerospace, pressure-vessel, fire-protection, or personal-protective use.